UNIMORE
UNIVERSITÀ DEGLI STUDI DI MODENA E REGGIO EMILIA

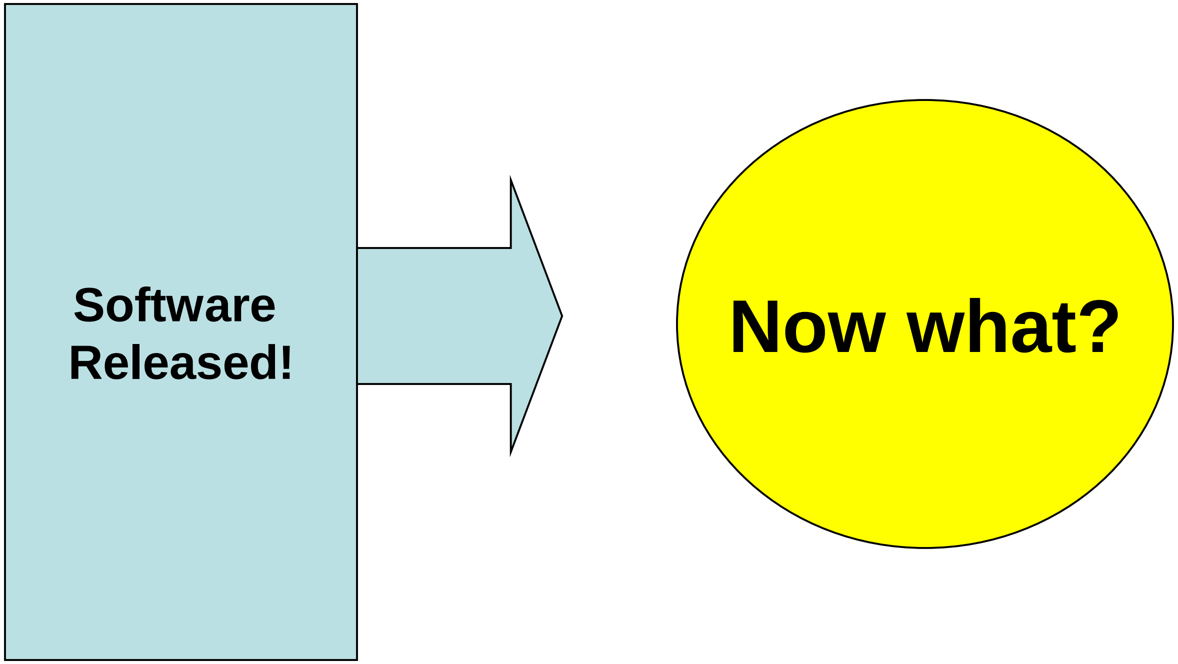
Software Support and Maintenance

Francesco Guerra
francesco.guerra@unimore.it

OBJECTIVES

- Describe post-release product support and customer service activities.
- Discuss product defect and non-defect support tasks.
- Analyze a sample customer support and service organization.
- Describe product-fix releases, which contain only product defect fixes, and maintenance releases, which contain small product enhancements and defect fixes.

Product released!



Customer Support

- Large software products typically require a **long and expensive development cycle.**
  - The **post-release support and maintenance phase** is **several times longer** than development.
- The ultimate success of the product depends heavily on **real-world user experience.**
  - Complex software often contains **defects at release**, despite extensive testing.
  - Resource and scheduling constraints may force early releases to **exclude some requirements**, leading to incremental delivery.
- Because the long-term viability of the product depends on effective support, it is crucial to prepare thoroughly for the post-release phase:
  - **Product Defect Support**
    - Identifying, diagnosing, prioritizing, and resolving defects reported by users.
    - Managing patches, updates, and hotfixes.
    - Ensuring proper communication and documentation.
  - **Product Non-Defect Support and Services**
    - Assisting users with configuration, installation, and usage questions.
    - Providing training, documentation, and usability support.
    - Handling enhancement requests and new feature proposals.

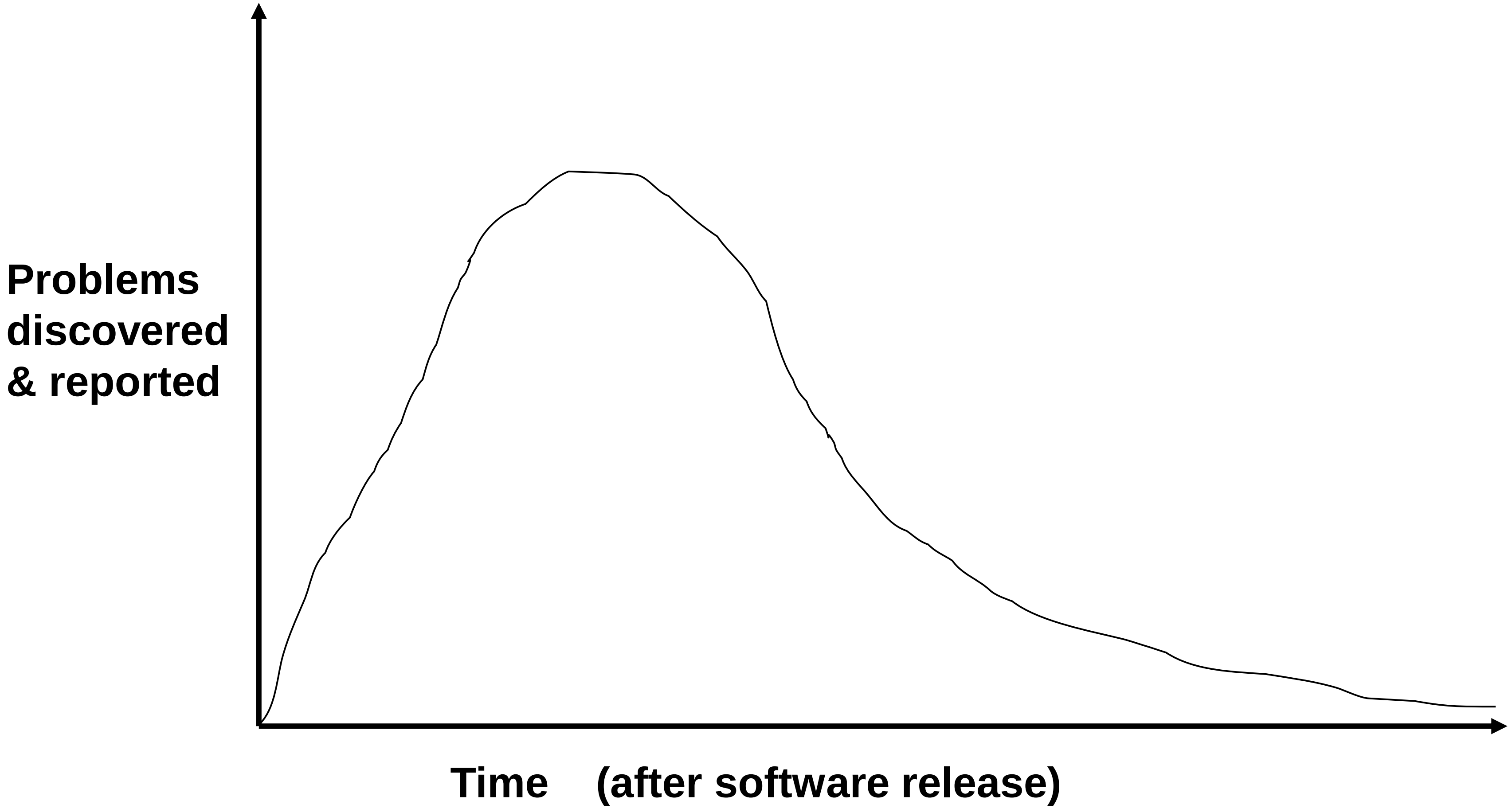
Customer Support

- Most customer support and service offerings **charge a fee**. Some providers use a **per-call pricing model**, where each support request has a cost.
- Paying customers expect:
  - **Professional-level services**
  - **Timely maintenance updates**
  - **High-quality assistance**
- **Support as a Consulting Practice**
  - Customer support in the form of a **consulting service** may include:
  - Product education and training
  - Mentoring and usage guidance
  - Product extensions or enhancements
  - Customizations tailored to specific customer needs

Customer Support

- **Special Case: Open Source Software**
  - For open-source products, support and customer service may be **problematic**.
  - The software is **free**, and therefore the creators or maintainers have **no legal obligation** to provide support.
  - Users may rely on:
    - Community forums
    - Volunteer maintainers
    - Paid commercial support (if offered by third parties)

User Problem Arrival Rate



User Problem Arrival Rate

- Sometimes a software product is **mistakenly assumed to be high quality** because the initial stream of problem reports is very low.
  - A low number of reports does **not necessarily** mean few defects.
  - It is essential to check the **actual usage rate** of the software.
- **Usage-Time Adjustment**
  - To correctly interpret the arrival rate of problems, we can normalize defect reports using **usage-time**.
  - **Usage-Time Unit Definition:**
    Usage month = (number of active users) × (months of usage)
  - This allows us to plot a more accurate **problem arrival curve** that reflects real exposure, not just calendar time.
- Accurately estimating the problem arrival rate is vital for:
  - Planning maintenance resources
  - Understanding product quality in real conditions
  - Predicting long-term support needs

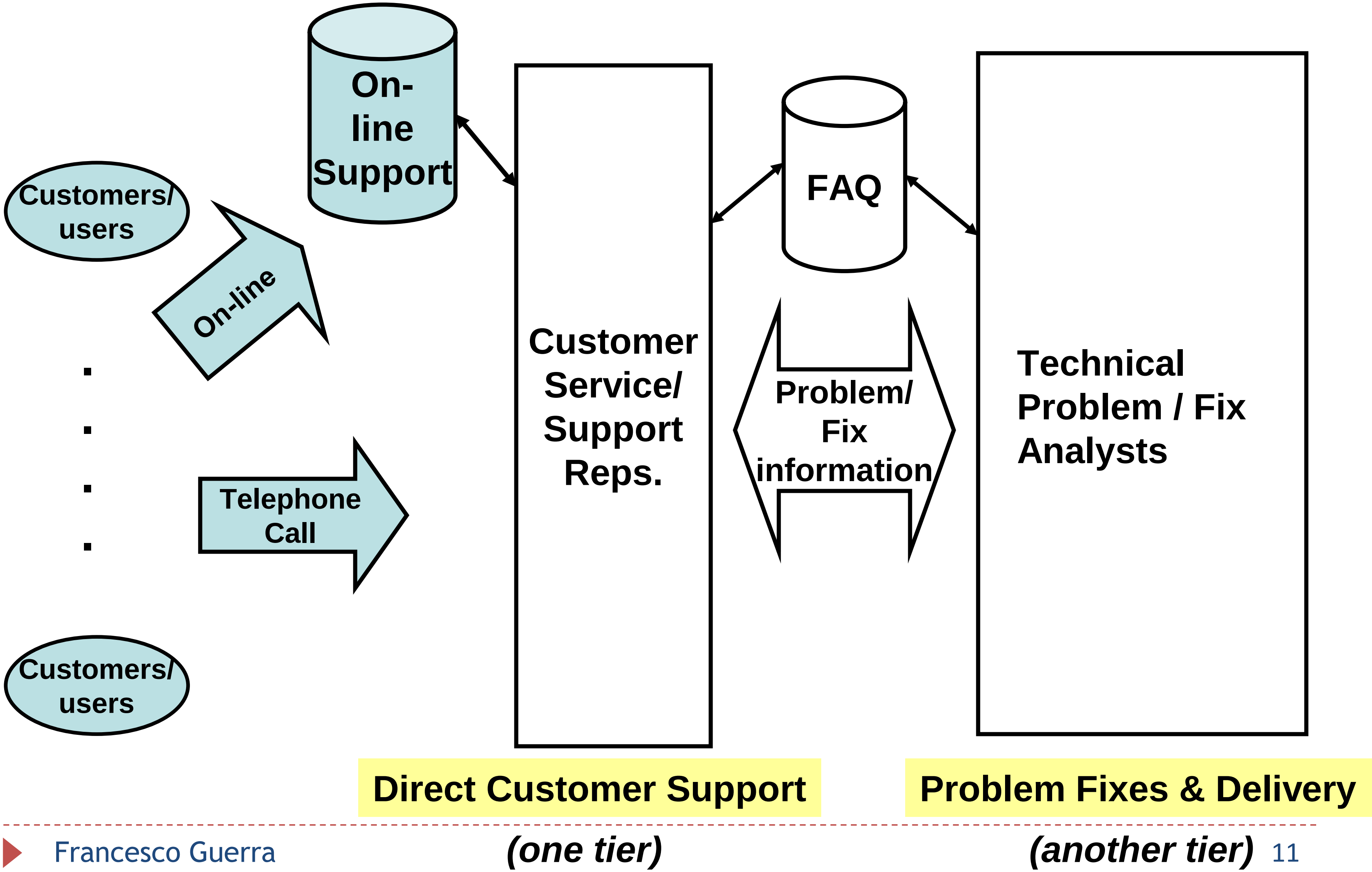
User Problem Arrival Rate

- **Preparing for Customer Support**
  - Preparation for customer support must start **before the software is released.**
  - Support personnel should receive **training on the product**, including features, and known issues.
  - Support staff should be involved in **testing activities**, to learn the system hands-on.
- **Knowledge Transfer to Support Teams**
  - A tester is **embedded ("seeded")** into the support team to serve as a mentor. This improves:
    - Knowledge transfer
    - Problem diagnosis skills
    - Communication between development and support
- **Estimating Support Workload**
  - Before release, organizations estimate:
    - Expected **support workload**
    - Number of **support personnel** required
    - Skills and training needed for effective problem resolution

User Problem Arrival Rate

- Usage and problem reports can be analyzed.
  - The data from the first months are used as a checkpoint to see if defect discovery projections are tracking to projection.
  - A renegotiation of support resources may be needed, if more problems are coming in than were originally projected.
- After a period time, the product can enter a sunset period and be taken off the market.
- The product sunset period usually go through the following sequence:
  1. Stop any product’s new feature or functional enhancement release.
  2. Fix only very high-severity problems related to the old product.
  3. Announce a new replacement product.
  4. Encourage existing users and customers to migrate to the new product.
  5. Notify users remaining on the old product of the planned termination of product support.
  6. Provide customers who want to remain with the old product possible vendors willing to continue the support.
  7. Terminate customer support on the planned date and withdraw the software product from the market.

Customer Interface and Call Management



Customer Interface and Call Management

- When a user encounters a problem, the approach is to **contact customer support**. The support representative then follows a structured sequence of activities:

1. **Customer Identification:** Verify that the user is a **qualified customer**, such as someone who has paid for the software or service subscription.
2. **Problem Description: Record a detailed description** of the problem.
3. **Initial Problem Search:** Check the **FAQ or knowledge base** to find similar problems and solutions.
4. **If the Problem Is Already Known**
   - **Solved problem:** Provide the documented solution.
   - **Known but unresolved problem:** Inform the customer of the **expected fix date** or update schedule.
5. **If the Problem Is New**
   - Provide a **work-around**, if one exists. If no work-around is available:
     - Agree with the customer on a **priority level** for the issue.
     - Give an **estimated fix time**.
6. **Reporting and Escalation:** Record the problem in the tracking system. Issue a **formal report** to the technical team responsible for fixing the problem.

Customer Interface and Call Management

- **Call Management** is the process of **handling customer support interactions** and ensuring that they are processed efficiently and smoothly.
  - Customer support representatives resolve only those problems that **do not require code fixes**.
  - Automated call management tools help supervisors collect **usage and performance information**.
- **Asynchronous Customer Support**
  - Support teams often maintain an **online service website**. Many activities normally performed by telephone are automated through this interface:
    - The reported problem is automatically checked against the **FAQ or knowledge base**. If a match is found, an **automatic response** is provided.
- **Handling New Problems**
  - If the issue is **new**, it is recorded and later reviewed by a customer service representative. Once analyzed, a response is sent to the customer, including:
    - The assigned **priority level**
    - The **estimated fix date** or update schedule
- **Escalation and Communication:** Many support organizations provide an **escalation facility** to handle dissatisfied or urgent customer cases.
  - Support or service websites may also publish: Product marketing announcementsm, Actual code fixes and patch information, Notifications about upcoming releases

Customer Interface and Call Management

- **Customer Expectations and Priority Handling**
- Experienced customers understand that the best way to receive **fast attention** is to have their problem assigned a **high priority level.**
- There is a direct relationship between **problem priority** and **problem-fix response time**:
  - Higher priority → faster response
  - Lower priority → slower response

- **Service Agreements and Customer Demands**
- Many experienced customers request specific **service-level agreements (SLAs)** *before* signing and paying for a support or service contract.
- These agreements often specify:
  - Maximum response time by priority level
  - Expected time to resolution
  - Availability and coverage conditions

Priority Level	Problem Category	Fix Response Time
1	Severe functional problem with nowork-around	As soon as possible
2	Severe functional problem but has work-around	1 – 2 weeks
3	Functional problem that has a work-around	3 – 4 weeks
4	Nice to have or to change	Next product release or earlier

Technical Problem/Fix

- When a problem is **too complex** or outside the expertise of the front-line support representative, it is escalated to a **technical problem/fix analyst.**
- A **formal change request (CR)** is created so that a **permanent fix** can be developed.
- **Fix Development Process**
  - Based on the change request, the technical analyst performs the following steps:
  - **Designs** the solution
  - **Codes** the fix
  - **Tests** the updated component
  - For **very high-priority issues**, the fix is packaged and delivered to customers **immediately**.
  - The fix is also included in the next **fix release**, a periodic collection of problem resolutions delivered to all customers.

Technical Problem/Fix

- **Role and Expertise of Technical Analysts**
  - Technical problem/fix analysts are typically **highly skilled engineers** with expertise in:
    - Software design
    - Coding
    - Testing
  - They generally **do not have direct support** from:
    - The original requirements analysts
    - The original product designers
  - Because of this, they must rely on:
    - **Deep knowledge of the product**
    - **Domain or industry expertise**
    - **Strong technical proficiency**
  - These skills allow them to diagnose and resolve issues independently.

Technical Problem/Fix

- The process of passing a customer problem from the service representative to the technical problem/fix analyst is :

1. Problem description, problem priority, and other related information is recorded in a **problem report** that is submitted to the problem-fix-and-delivery group.
2. The problem-fix-and-delivery group will explore and analyze the problem, including the reproduction of the problem.
3. The problem-fix-and-delivery group either accepts or rejects the problem.
4. If the problem is rejected, the direct customer support group is immediately notified; if the problem is accepted, then a change request is generated and the problem enters a fix cycle of design, code, and test.
5. The fix may be individually packaged and released immediately to the customer or the fix may be integrated into a fix release package.
6. The FAQ database is updated to reflect the status of all the problems so that the customer support/service representatives may advise the customers on the problem resolution status.

Technical Problem/Fix

- The **design-code-test cycle** used for problem fixes is fundamentally the same as in the original development process.
- However, there is a significant **risk**:
  - Because many fixes involve only **one or two lines of code**, developers may be tempted to implement quick patches without following proper engineering discipline.
- For organizations managing **multiple releases** across **multiple countries**, it is imperative to maintain full engineering discipline:
  - Requirements, design, code, and test **documentation must be updated** for every fix.
  - A fix applied in one place often needs to be **propagated to other releases** and **localized versions**.
  - Without documentation and traceability, fixes may be inconsistently applied, causing future regressions.

Technical Problem/Fix

- **Increasing Need for Measurement and Control**
  - Customers increasingly demand **Service Level Agreements (SLAs)**, often with **penalty clauses** for missing response or fix deadlines.
  - This trend raises awareness of the need for:
    - Reliable and measurable support processes
    - Accurate fix-time tracking
    - Consistent documentation of problem-fixes
- **Handling Critical or Life-Threatening Situations**
  - In critical environments (e.g., medical, transportation, industrial systems), the support process differs from normal operations:
    - Support personnel may need to be **physically on-site** to diagnose and fix the problem immediately.
    - Local fixes made directly in the customer's environment must still be:
      - **Documented** properly
      - **Integrated back** into the general fix release
      - Shared with all customers who might be affected
  - This ensures consistency, safety, and regulatory compliance.

Delivery and Fix Installs

- **Regular and periodic fix releases** are on a quarterly or semiannual cycle.
  - They contain only those modules and code that are affected by the problem-fixes.
    - The fix code is packaged in a sequential order.
    - The fix releases are sequential, and the customers are expected to install the fix releases in sequential order.
    - All fix releases are made available to all the customers that have a service contract.
    - These fix releases, sometimes called patches, are downloadable from an online site or are shipped to the customers on CDs.
- Not all customers will apply (or install) the fix releases immediately.
  - Customers who have achieved a "working" state and do not want to risk their working software with any new fix releases.
  - They may, however, encounter a problem later and want a particular problem-fix.
  - Such a retroactive application of all the fix releases can be very time consuming and frustrating when the customer wants only one problem fixed or a single patch.
  - To keep the support service effort at a reasonable level, customers are always encouraged to apply the fix release as soon as possible after it becomes available.

Delivery and Fix Installs

- High-priority emergency fixes may be delivered to the customer whose system may be down and whose business is suspended.
  - The customer will immediately apply the fix upon receiving it.
  - The customer's system is brought up and business resumes.
  - When the next regular fix release is made available, they can perform a partial install of the fixes in the fix release, figuring that they already have the emergency fix applied.
  - Some of the fixes in the accumulated fix release may impact the emergency fix that was already applied.
    - The customers may have to back out that emergency fix first and then apply the complete fix release.
- When a new fix release is available, an information document that contains **the list of problems and corresponding fixes is included along with advice to customers who may be at different release**, version, or fix release

Delivery and Fix Installs

- When software applications are intertwined, a fix release of one must be co-applied with a fix release of another software product.
  - For example, an operating system fix release may be related to a fix release of a database system and of a network system.
  - The three fix releases are co-requisites of each other and must be applied together for all the fixes to work.

Change Control

- A large customer service organization must have a well-established change control.
- It must control all the changes, whether the change originated from a customer problem or a planned small, incremental, feature enhancement.
- The scheduling of changes based on the number of anticipated problems and change requests is usually resource gated.
- Change control is a process that manages the flow of the following activities:
  - Origination of a change request
  - Approval of a change request
  - Acting on a change request
  - Tracking and closing a change request
- Along with the activity flow, there is also certain information that must be passed through the flow. The information relevant to a change request is usually captured in a change request form

Sample Maintenance Change Request Form

Change Request # :______________ **Request Date:**______________

Requestor Name:________________

Request Status: ***Accepted-Date:***________________
***Rejected-Date*:**________________
***Processing Start-Date* :**________
Completion-Date:______________

Requestor Priority: ***High, Medium, Low***

Brief Change Request Description:____________________________________
____
____
____

Areas Impacted by the Change Request :________________________________
____
____
____

Estimated Effort: ____________ **Inclusion in Maintenance Rel.#:** ____________

Change Control (2)

- There are more comprehensive change request forms for organizations that desire to manage the change request more thoroughly.
  - The estimated effort field may be modified into two fields that capture the initial effort estimate and the actual effort expended.
  - Also, there may be a field that shows the actual modules, database, and so on, that were impacted by the change request.
  - However, it is important that the form is not made more complex than necessary. There must be a good reason to have a field included in the form.
- Many software organizations perform extensive studies over these change request forms to better understand their product shortcomings, customer needs, and future product directions.

UNIMORE
UNIVERSITÀ DEGLI STUDI DI MODENA E REGGIO EMILIA

References

- Frank Tsui, Orlando Karam, Barbara Bernal, Software engineering (Third Edition), Jones & Bartlett Learning